# Bihar Engineering University, Patna

## B.Tech 6th Semester Examination, 2024

*Course : B.Tech*
*Code : 101602*

*Subject: Construction Engineering & Management*

*Time: 03 Hours*
*Full Marks: 70*

***Instructions:-***

*(i) The marks are indicated in the right-hand margin.*
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

**Q.1 *Choose the correct answer the following (Any seven question only)*** **[2 x 7 = 14]**

(a) A Project charter is prepared and issued by
(i) Stakeholders
(ii) Project Manager
(iii) Project sponsor
(iv) Project management office

(b) The best way to resolve a conflicts in a project is by
(i) Collaborative
(ii) Confronting
(iii) Compromising
(iv) Smoothing

(c) You as a Project manager want to do demand forecast for your new product. Which type of forecast will you do to analyse the market?
(i) Qualitative
(ii) Quantitative
(iii) Any one of the above
(iv) Moving average method

(d) Which one of the following statements is not correct?
(i) PERT is probabilistic whereas CPM is deterministic.
(ii) In PERT slack on various events in calculation whereas in CPM floats are calculated.
(iii) Critical path in a network is the path on which events have no slack.
(iv) More than four dummy activities cannot be used in a PERT network.

(e) Which among the following is faster method of finding activity duration?
(i) Analogous method
(ii) Bottom up method
(iii) Can't say
(iv) Both are very time consuming

(f) In a project to create a new mobile phone your sponsor asks you that the launch date is brought forward. What will be its impact?
(i) cost increases
(ii) cost decreases
(iii) scope increases
(iv) cost remains unaffected

(g) A Positive cost variance and negative schedule variance means
(i) Project is over budget and ahead of schedule
(ii) Project is under budget and behind of schedule
(iii) Project is over budget & behind schedule
(iv) Project is under budget and ahead of schedule

(h) Pick up the incorrect statement from the following
(i) An activity of a project is denoted by an arrow on the network.
(ii) The tail of the arrow indicates the start of the activity
(iii) The head of the arrow indicates the end of the activity
(iv) the arrows are drawn to scale from left to right

(i) Milestone charts were invented in the year of
(i) 1910
(ii) 1920
(iii) 1930
(iv) 1940



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(j) A dummy activity
(i) is artificially introduced
(ii) is represented by a dotted line
(iii) does not consume time
(iv) All of the above

**Q.2** Differentiate between the following:
(a) Activities on Arrows (AOA) and activities on Nodes (AON) network diagrams. [7]
(b) CPM and PERT techniques [7]

**Q.3** (a) What are the various stages in construction? Describe each stage briefly with special reference to the economy that can be exercised in each. [7]
(b) What are the requirements for an ideal formwork for concreting? [7]

**Q.4** A project consists of some activities as given below : [14]

| Job Activity | Immediate Predecessors | Activity duration |
|---|---|---|
| A | - | 4 |
| C | A | 6 |
| D | B | 4 |
| E | B | 5 |
| F | C | 3 |
| H | D | 11 |
| I | E,F | 9 |
| J | G | 6 |
| K | I,J,K | 0 |

Draw an arrow diagram of the project. Also find EST, LST, FFT and LFT, project duration and total float.

**Q.5** (a) What are the advantages and disadvantages of a dragline over a power shovel. Describe the basic parts of a dragline by a sketch. [7]
(b) Discuss the various means for hauling the materials. [7]

**Q.6** (a) Describe the role of construction planning in the management of construction industries. [7]
(b) What is the significance of preparatory work prior to the execution of construction Projects? [7]

**Q.7** (a) Discuss the importance of project monitoring. [7]
(b) Define specifications. What are the technical specifications and how these are enforced in the field? [7]

**Q.8** (a) Define the term contractor. What are the essential qualifications of a contractor? [7]
(b) Describe five main provisions of the Contract Labour Act, 1970. [7]

**Q.9** (a) Define and explain the direct and indirect cost of an accident. [7]
(b) Enumerate the factors which affect the analysis of rate of any item. [7]

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